COURSE NAME

DISCRETE MATHEMATICS

CSC 1204

(UNDERGRADUATE)

CHAPTER 11

MATHEMATICAL INDUCTION

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MATHEMATICAL INDUCTION

- Suppose that we have an infinite ladder, and we want to know whether we can reach every step on this ladder.
- We know two things –
  1) We can reach the first step of the ladder
  2) If we can reach a particular step of the ladder, then we can reach the next step
- Can we conclude that we can reach every step?
- By (1), we know that we can reach the first rung of the ladder. Moreover, because we can reach the first rung, by (2), we can also reach the second rung; it is the next rung after the first rung.
- Applying (2) again, because we can reach the second step, we can also reach the third step. Continuing in this way, we can show that we can reach the fourth step, the fifth step, and so on.
- For example, after 100 uses of (2), we know that we can reach the 101st rung

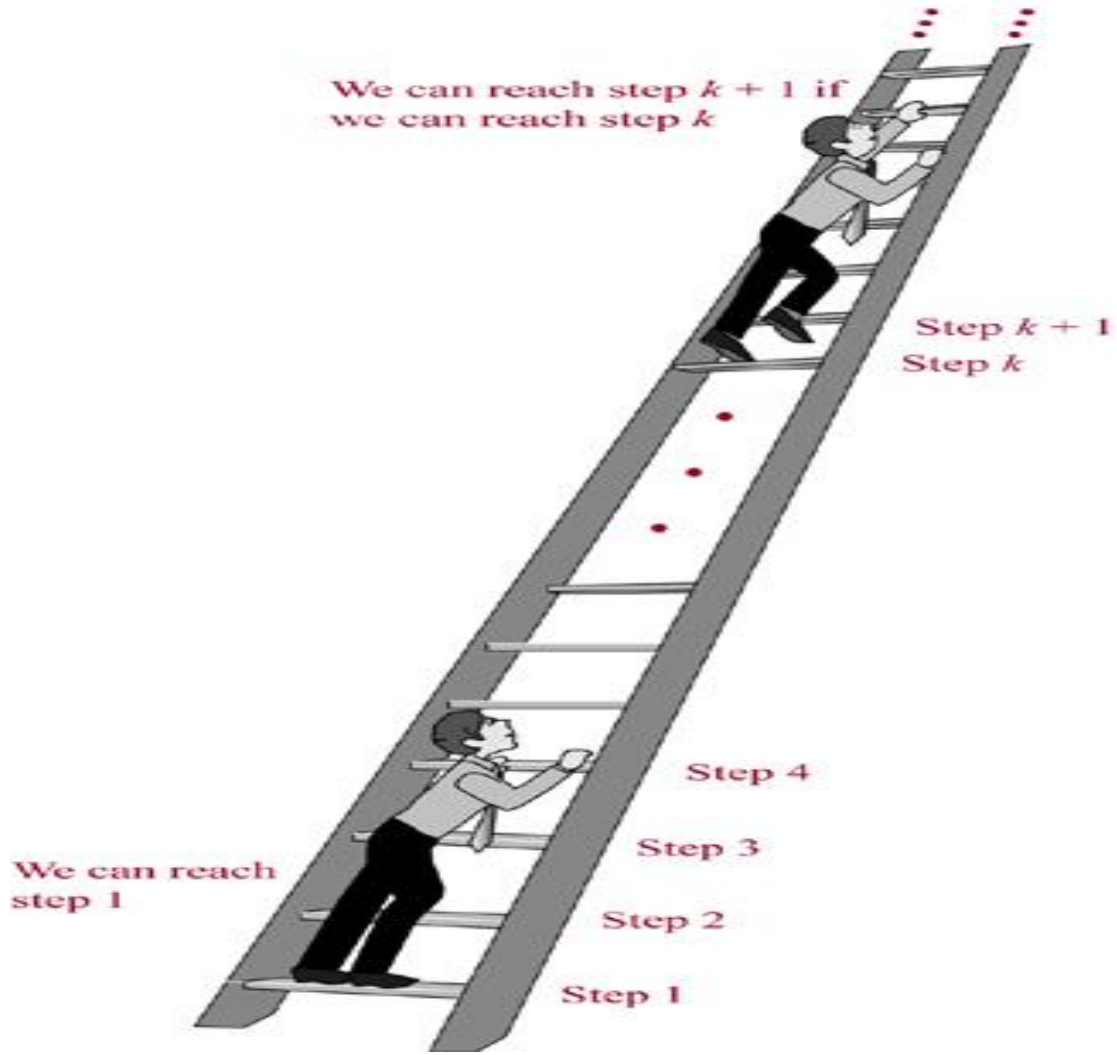
MATHEMATICAL INDUCTION

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Q. Can we conclude that we are able to reach every rung of this infinite ladder?

Answer: Yes.

- We can verify using an important proof technique called mathematical induction
- We can show that $P(n)$ is true for every positive integer n, where $P(n)$ is the statement that we can reach the nth rung of the ladder.



MATHEMATICAL INDUCTION

- Mathematical induction is an extremely important proof technique that can be used to prove assertions of this type.
- Mathematical induction is used extensively to prove results about a large variety of discrete objects.
- For example, it is used to prove results about the complexity of algorithms, the correctness of certain types of computer programs, theorems about graphs and trees, as well as a wide range of identities and inequalities
- Mathematical induction can be used only to prove results obtained in some other way. It is *not* a tool for discovering formulae or theorems.
- The principle of mathematical induction is a useful tool for proving that a certain predicate is true for all natural numbers
- It cannot be used to discover theorems, but only to prove them

PRINCIPLE OF MATHEMATICAL INDUCTION

- To prove that P(n) is true for all positive integers n, where P(n) is a propositional function, we complete two steps:

  1) Basis step: we verify that P(1) is true.

  2) Inductive step : we show that the conditional statement P(k) $\rightarrow$ P(k+1) is true for all positive integers k

     Inductive hypothesis: P(k) is true

- [$P(1) \wedge \forall k(P(k) \rightarrow P(k+1))] \rightarrow \forall n\, P(n)$

PRINCIPLE OF MATHEMATICAL INDUCTION

Q: Show that if n is a positive integer, then $1 + 2 + \ldots\ldots + n = n(n+1)/2$

Solution: Let *P(n)* be the proposition that the sum of the first n positive integers is n(n+1)/2. We must do two things to prove that *P(n)* is true for n = 1, 2, 3, ….. Namely, we must show that P(1) is true and that the conditional statement P(k) implies P(k+1) is true for k = 1, 2, 3,

- Basis step: P(1) is true, because $1 = 1(1+1)/2$
- Inductive step: For the inductive hypothesis, we assume that P(k) holds for an arbitrary positive integer k. That is we assume that $1 + 2 + \ldots..+ k = k(k+1)/2$ (Inductive hypothesis)

PRINCIPLE OF MATHEMATICAL INDUCTION

- Under this assumption, it must be shown that *P(k+1)* is true, namely, that

- $1 + 2 + \ldots\ldots + k + (k+1) = (k+1)\,[(k+1)+1]\,/2 = (k+1)(k+2)/2$ is also true. $[(k+1)+1 = k+2)]$
- Adding (k+1) to both sides of the equation in P(k), we obtain

$$1 + 2 + \ldots\ldots + k + (k+1) = k(k+1)/2 + (k+1)$$

$$= [\, k(k+1) + 2(k+1) \,] / 2$$

[multiple by 2 with last part to make whole R.H.S. divided by 2]

$$= (k+1)(k+2)/2$$

- This last equation shows that P(k+1) is true under the assumption that P(k) is true. This completes the inductive step.
- We have completed the basis step and inductive step, so by mathematical induction we know that P(n) is true for all positive integers n. That is, we have proven that $1 + 2 + \ldots + n = n(n+1)/2$ for all positive integers n.

REFERENCES

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- *SCHAUM'S ouTlines Discrete Mathematics(2nd edition)*, by Seymour Lipschutz, Marc Lipson
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